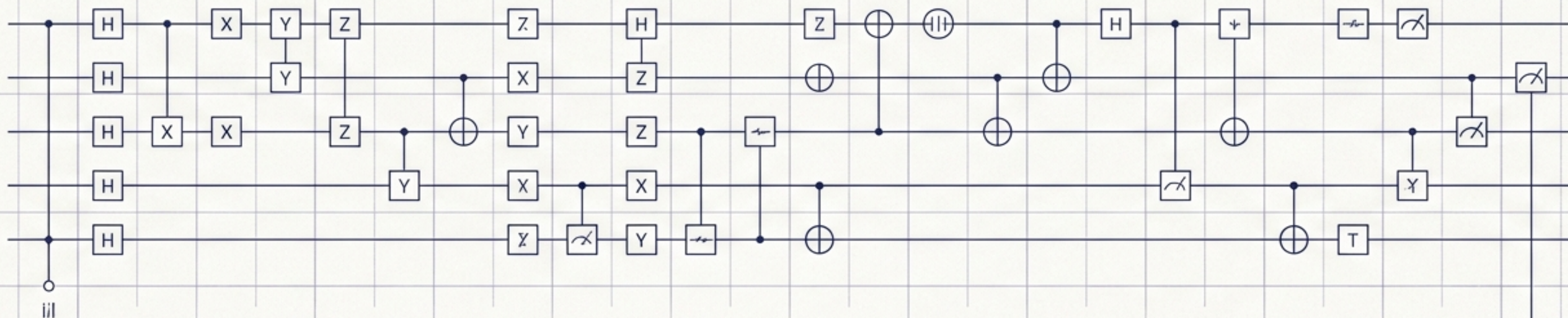
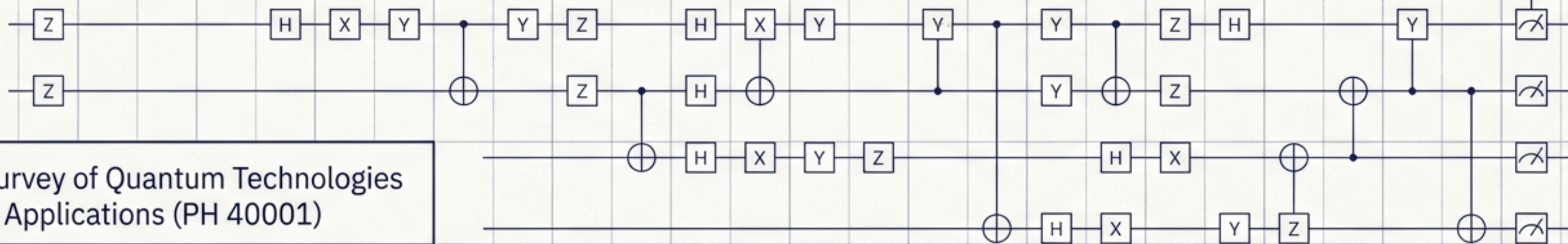




# Architecting Quantum States

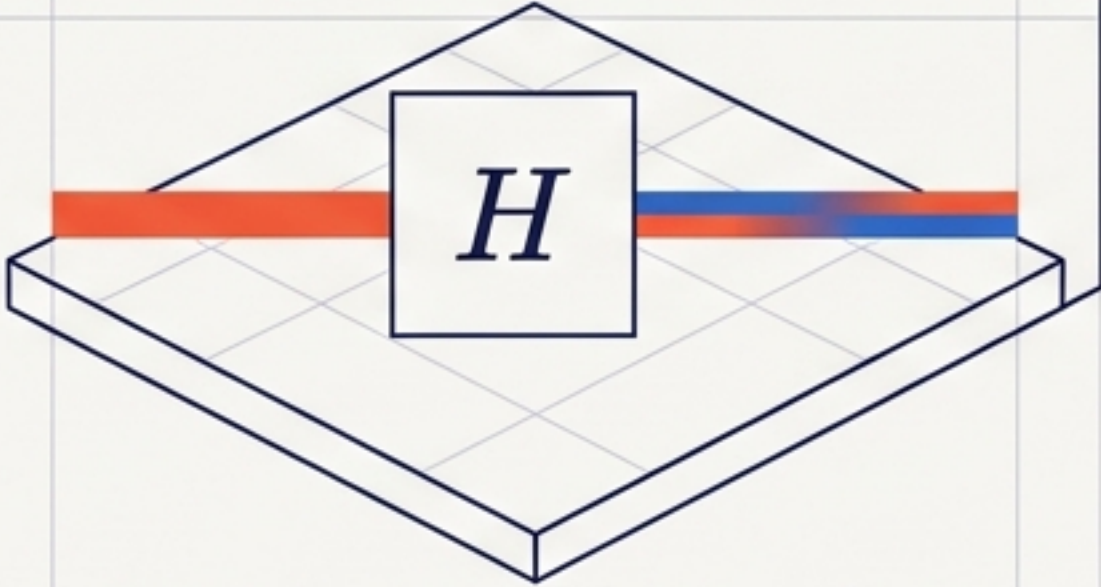
## From Single-Qubit Gates to Entanglement Machines



Survey of Quantum Technologies  
& Applications (PH 40001)

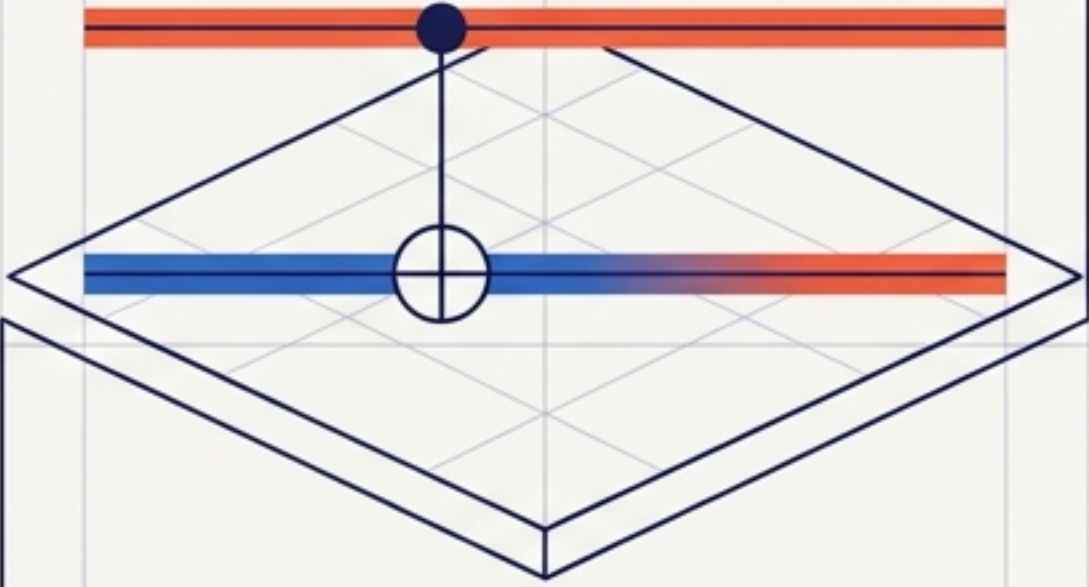
Lecture 5

# The Building Blocks of Quantum Computation



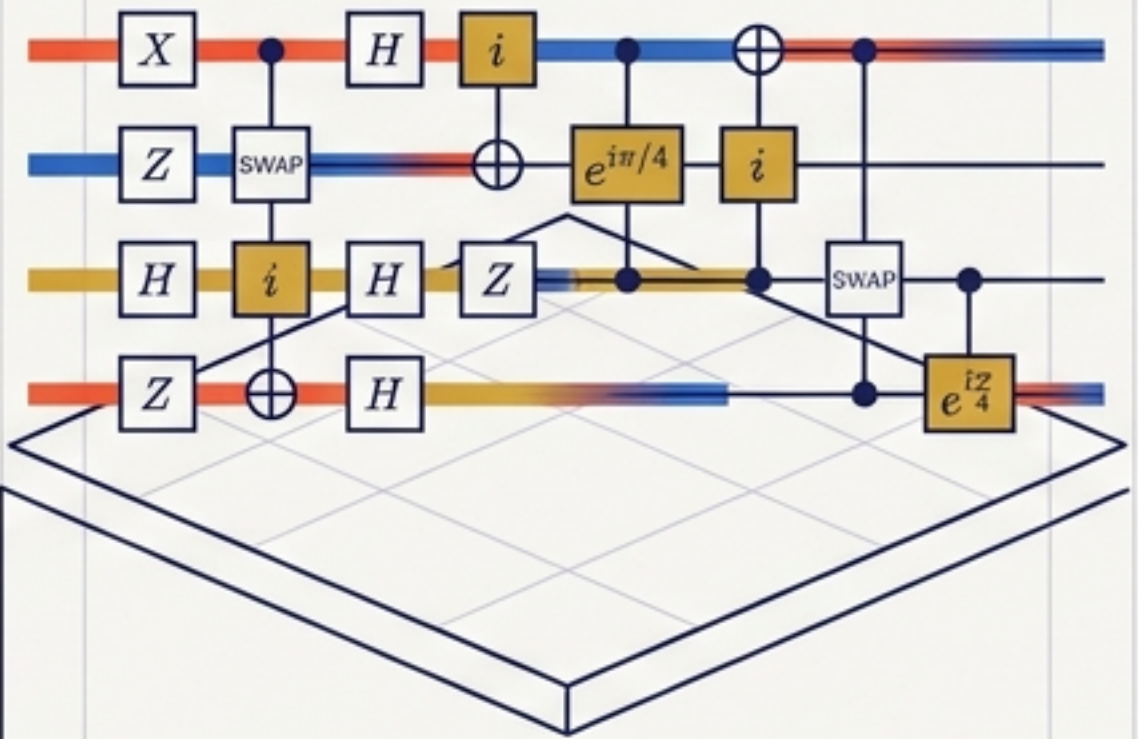
## 1. Atoms (Single-Qubit Gates)

Shaping individual states (H, X, Z, S, T).



## 2. Molecules (Two-Qubit Interactions)

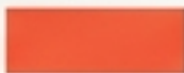
Conditional logic and bonding (CX, SWAP).



## 3. Machines (Quantum Architectures)

Functional outputs (Adders & Entangled Pairs).

State Legend:

|                                                                                       |                   |                                                                                       |                   |                                                                                       |                      |
|---------------------------------------------------------------------------------------|-------------------|---------------------------------------------------------------------------------------|-------------------|---------------------------------------------------------------------------------------|----------------------|
|  | $ 0\rangle$ State |  | $ 1\rangle$ State |  | Phase Transformation |
|---------------------------------------------------------------------------------------|-------------------|---------------------------------------------------------------------------------------|-------------------|---------------------------------------------------------------------------------------|----------------------|

# Diagnostic Matrix: Single-Qubit Gates (Atoms)

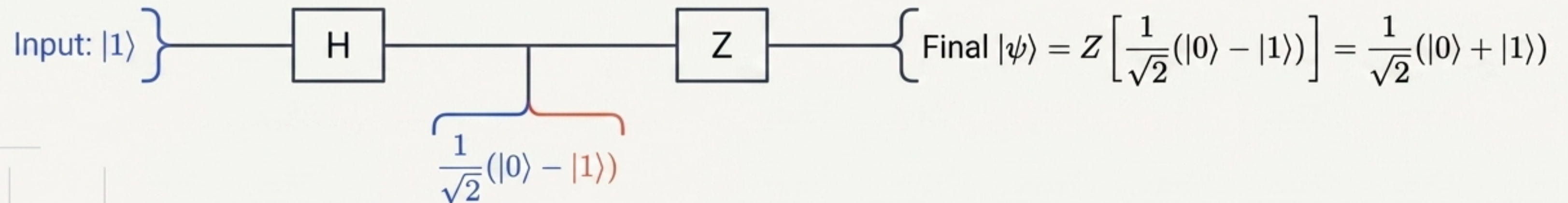
| Gate         | Circuit Symbol                                                                       | Action on $ 0\rangle$                       | Action on $ 1\rangle$                                                   | Layman's Function     |
|--------------|--------------------------------------------------------------------------------------|---------------------------------------------|-------------------------------------------------------------------------|-----------------------|
| Hadamard (H) |    | $\frac{1}{\sqrt{2}}( 0\rangle +  1\rangle)$ | $\frac{1}{\sqrt{2}}( 0\rangle -  1\rangle)$                             | Creates Superposition |
| Pauli-X (X)  |    | $ 1\rangle$                                 | $ 0\rangle$                                                             | Bit Flip              |
| Pauli-Z (Z)  |  | $ 0\rangle$                                 | $- 1\rangle$                                                            | Phase Flip            |
| Phase (S)    |  | $ 0\rangle$                                 | $i 1\rangle$                                                            | 90° Phase Shift       |
| T-gate (T)   |  | $ 0\rangle$                                 | $e^{\frac{i\pi}{4}} 1\rangle \rightarrow \frac{1+i}{\sqrt{2}} 1\rangle$ | 45° Phase Shift       |

# State Evolution Tracking: 2-Gate Cascades

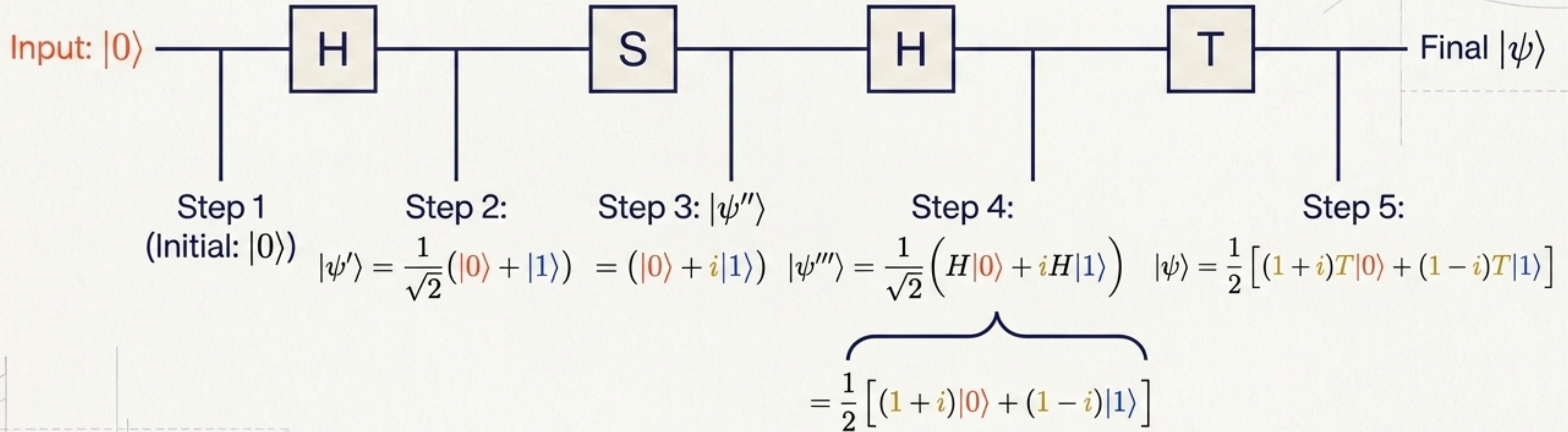


## Key Insight

Applying different gate sequences to different initial states can collapse into identical final quantum configurations.



# State Evolution Tracking: 4-Gate Cascade

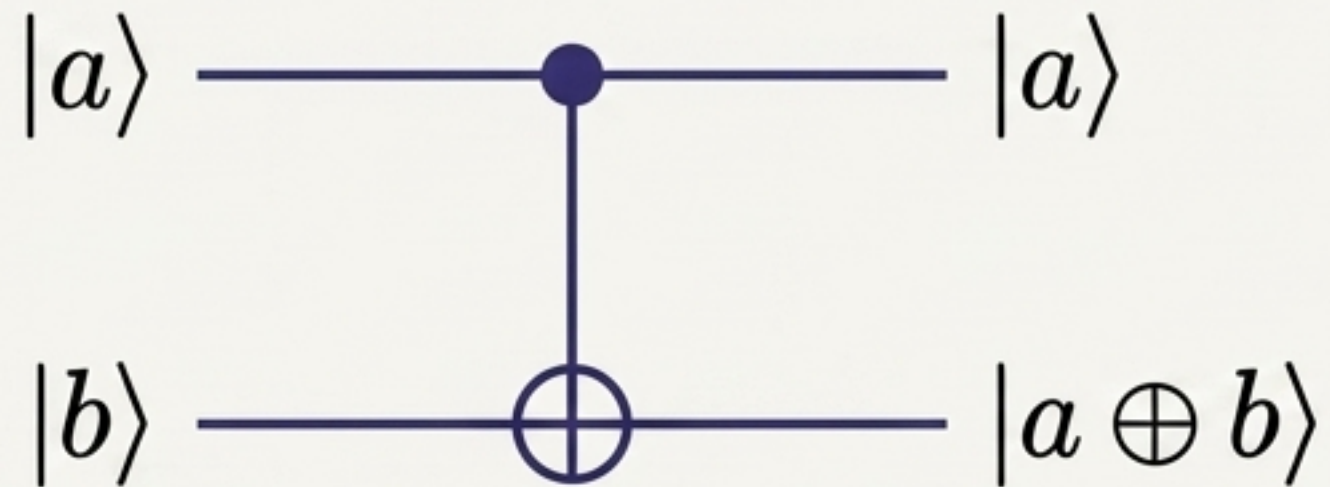


Final Equation Plate

Simplified Final Result:

$$|\psi\rangle = \frac{(1+i)}{2}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

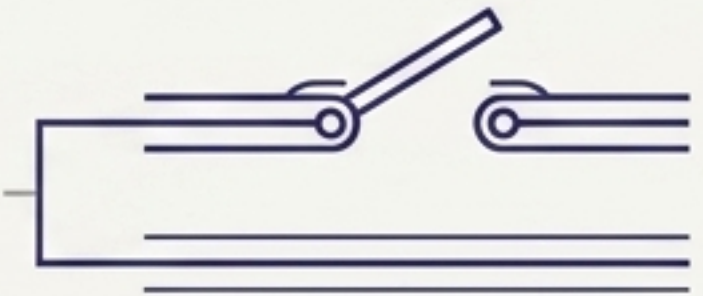
## Matrix &amp; Schematic



$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

## Physical Switch Metaphor

If Control  $|a\rangle = |0\rangle$



Switch is OPEN. Target  $|b\rangle$  passes through untouched.

$$\begin{aligned} |00\rangle &\rightarrow |00\rangle \\ |01\rangle &\rightarrow |01\rangle \end{aligned}$$

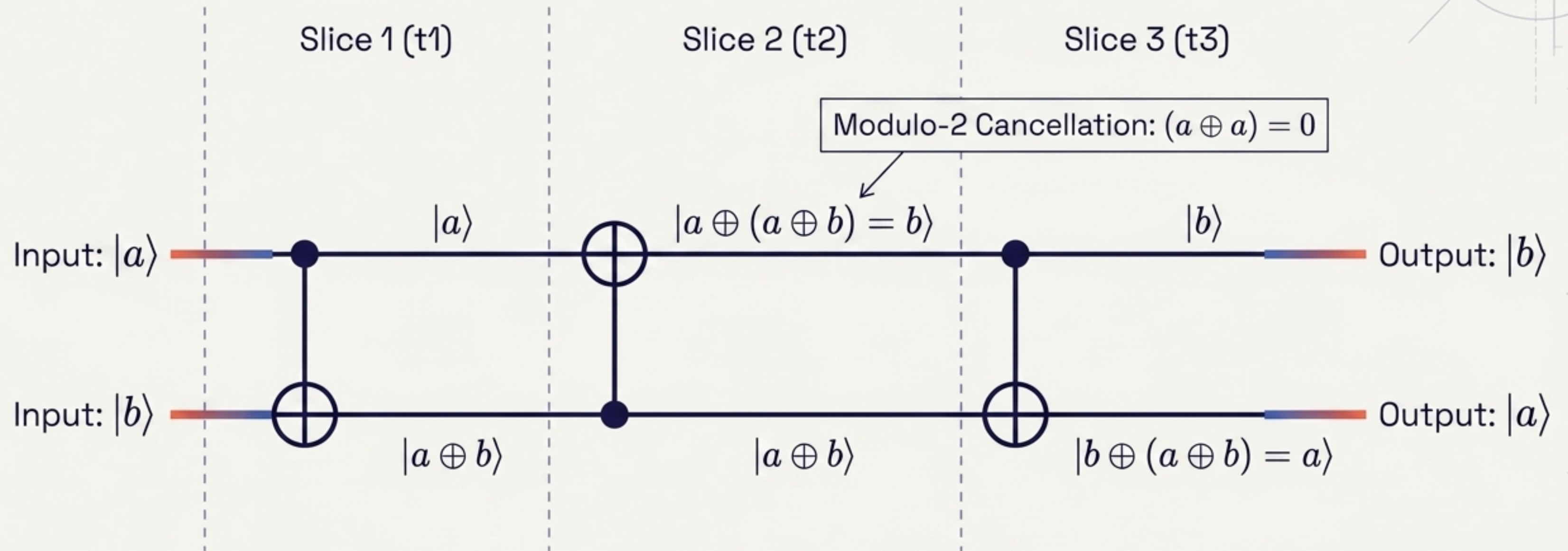
If Control  $|a\rangle = |1\rangle$



Switch ACTIVATES. Target  $|b\rangle$  hits a virtual X-gate and flips.

$$\begin{aligned} |10\rangle &\rightarrow |11\rangle \\ |11\rangle &\rightarrow |10\rangle \end{aligned}$$

# Anatomy of the SWAP Gate



Takeaway: 3 consecutive CX gates with alternating orientations perfectly transpose quantum states without physical wire crossover.

# Classical Emulation: The Quantum Half-Adder

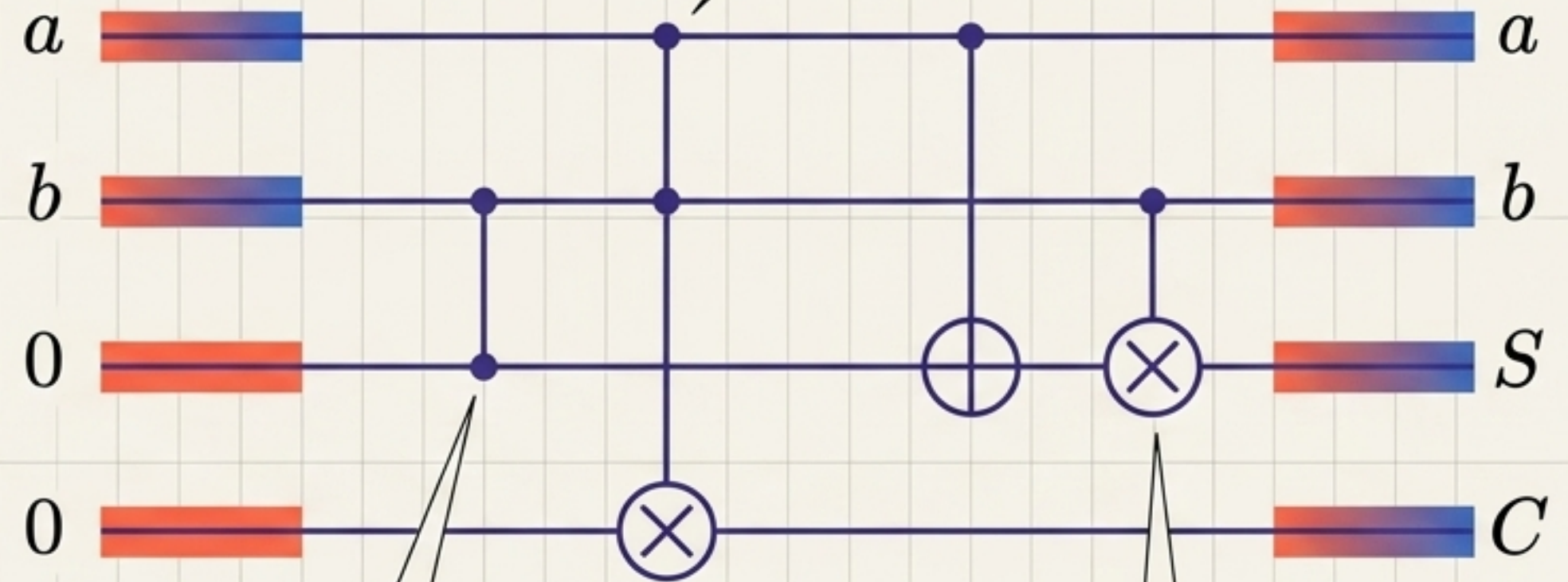
## Classical Boolean Logic

| $a$ | $b$ | $S$ | $C$ |
|-----|-----|-----|-----|
| 0   | 0   | 0   | 0   |
| 0   | 1   | 1   | 0   |
| 1   | 0   | 1   | 0   |
| 1   | 1   | 0   | 1   |

$$\text{Sum } (S) = a \oplus b \text{ (XOR)}$$

$$\text{Carry } (C) = a \cdot b \text{ (AND)}$$

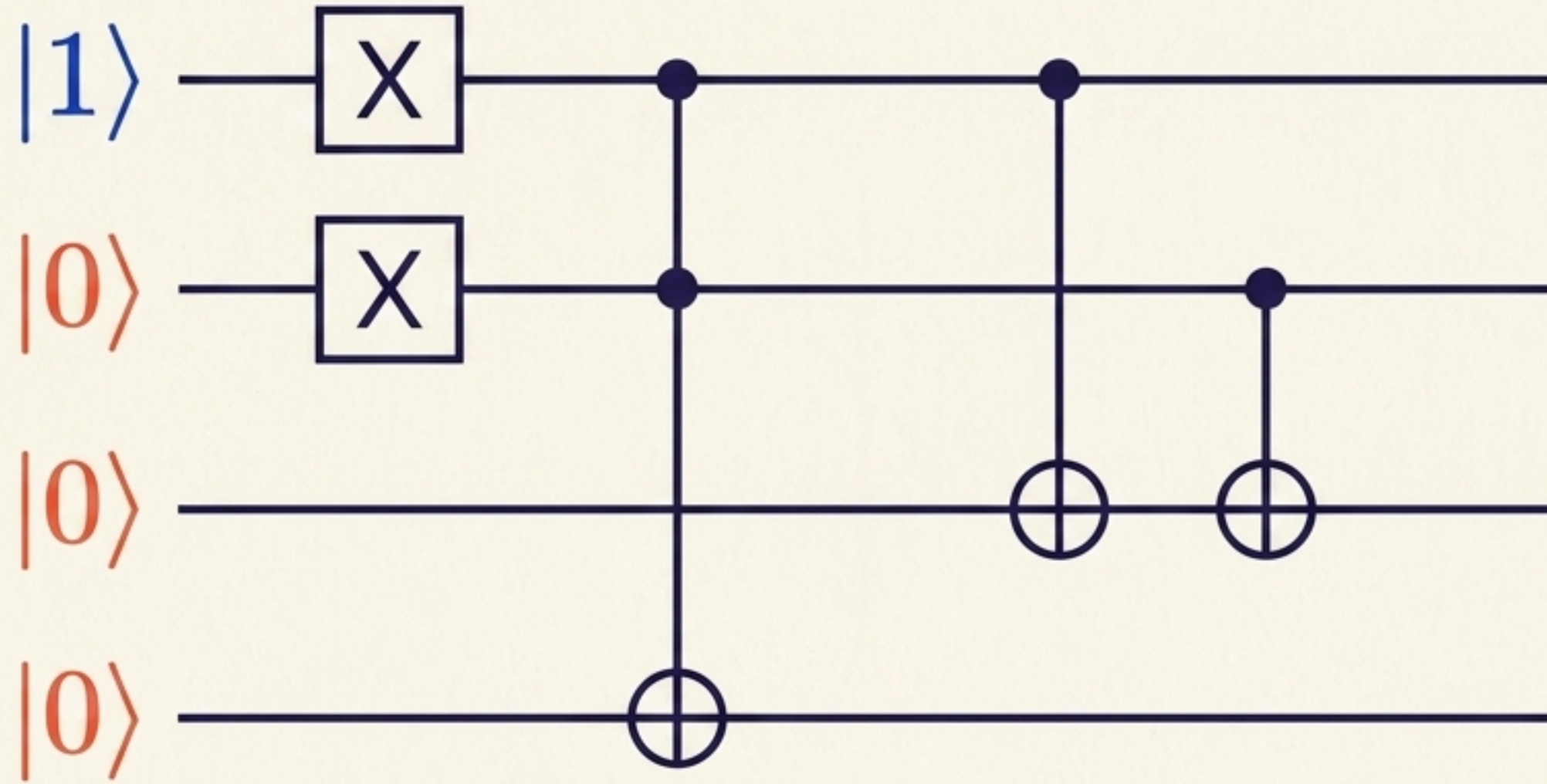
## 4-Wire Quantum Mapping



The CCX (Toffoli) gate acts as the quantum equivalent of a classical AND gate for the Carry bit ( $C = ab$ ).

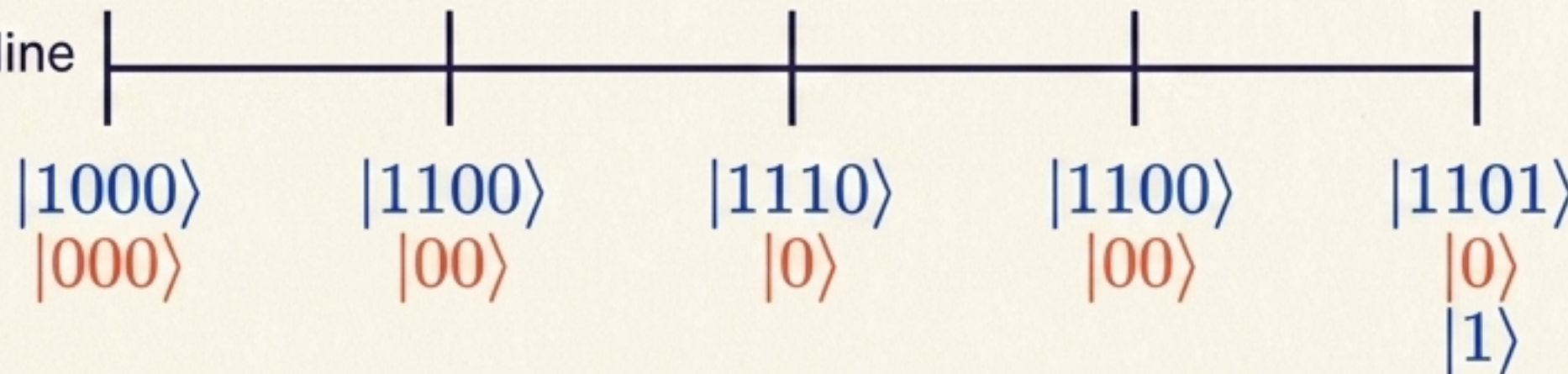
The cascaded CX gates act as the XOR function for the Sum bit ( $S = a \oplus b$ ).

# Half-Adder State Tracking: Practical Debugging



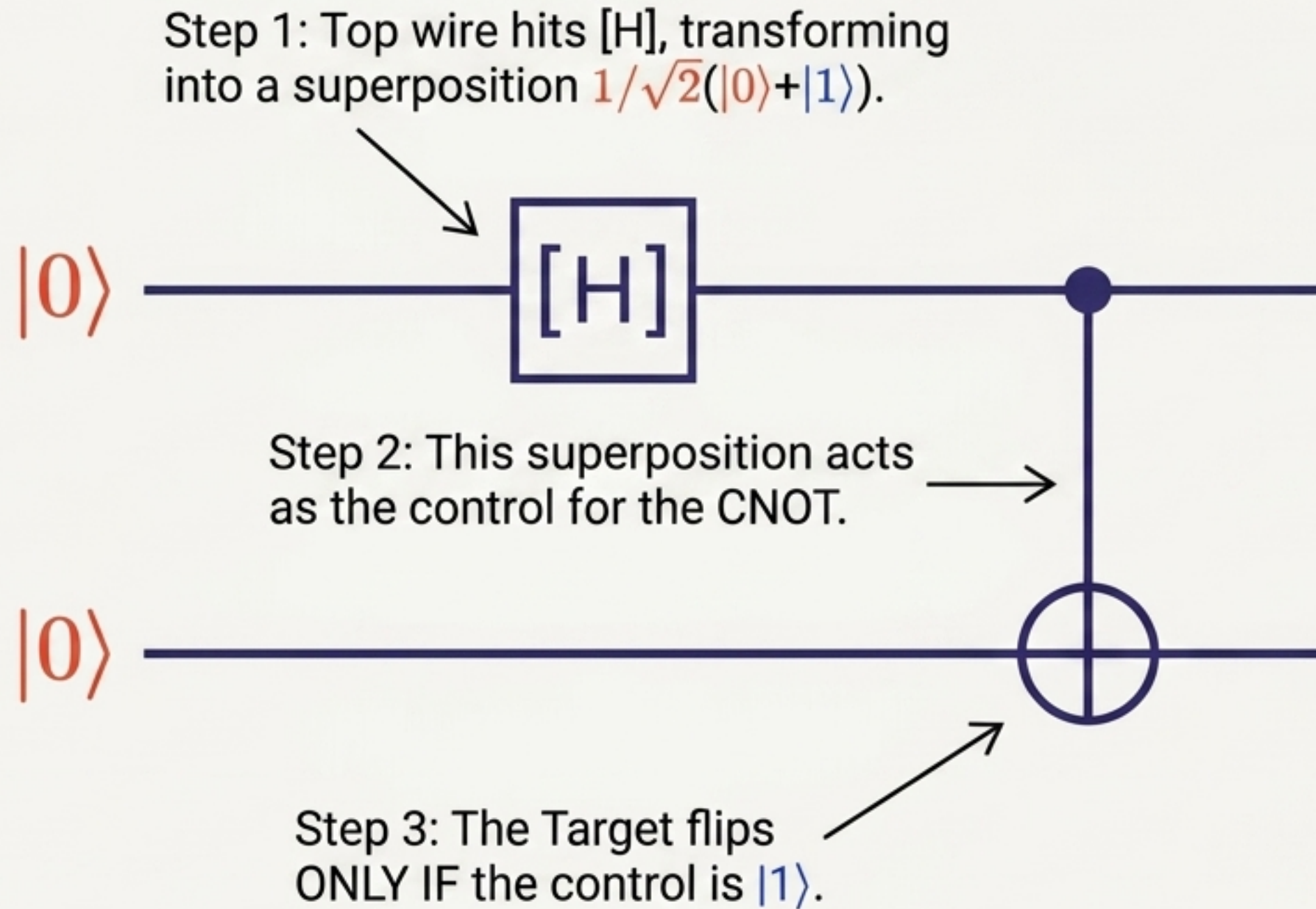
State Tracking Timeline

Input State  
Vermilion



Final output state yields  
Sum = 1, Carry = 0 for  
inputs 1 and 0, executing  
perfect arithmetic via  
reversible quantum logic.

# Quantum Supremacy: Generating Entanglement



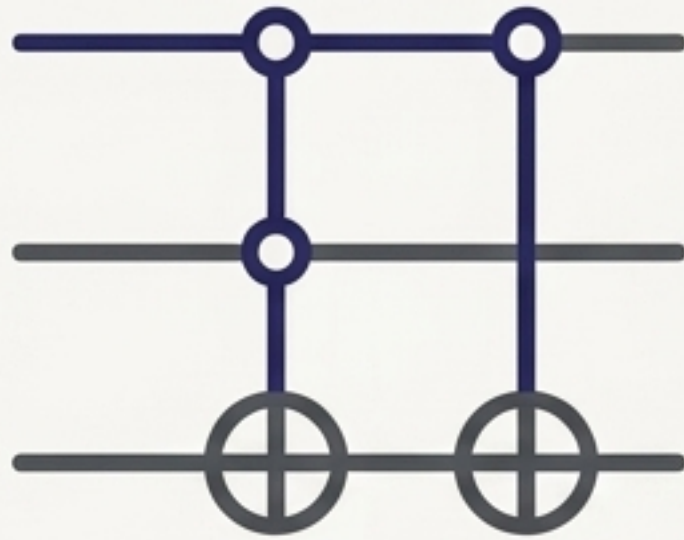
$$|\psi\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

The EPR State: The qubits are now perfectly correlated. Measuring one instantaneously determines the state of the other—a phenomenon with no classical equivalent.

# Diagnostic Matrix: The Four Bell States

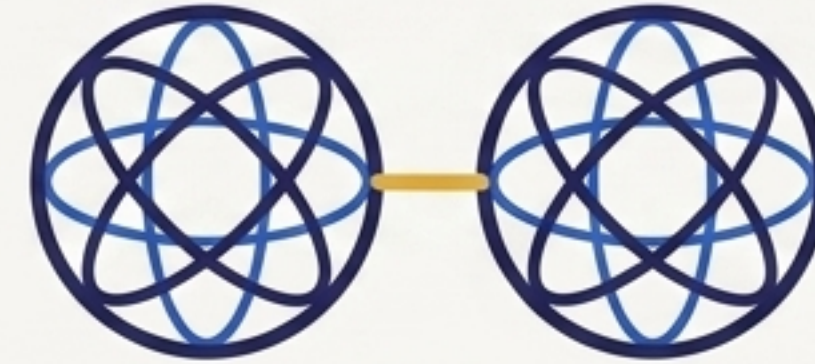
| Input State             | The Circuit                                                                           | Mathematical Expansion                               | Final Bell State (EPR Pair)                                       |
|-------------------------|---------------------------------------------------------------------------------------|------------------------------------------------------|-------------------------------------------------------------------|
| $ 00\rangle$            |    | $\frac{1}{\sqrt{2}}( 0\rangle +  1\rangle) 0\rangle$ | $ \psi_{00}\rangle = \frac{1}{\sqrt{2}}( 00\rangle +  11\rangle)$ |
| $ 01\rangle,  1\rangle$ |   | $\frac{1}{\sqrt{2}}( 0\rangle +  1\rangle) 1\rangle$ | $ \psi_{01}\rangle = \frac{1}{\sqrt{2}}( 01\rangle +  10\rangle)$ |
| $ 10\rangle,  0\rangle$ |  | $\frac{1}{\sqrt{2}}( 0\rangle -  1\rangle) 0\rangle$ | $ \psi_{10}\rangle = \frac{1}{\sqrt{2}}( 00\rangle -  11\rangle)$ |
| $ 11\rangle$            |  | $\frac{1}{\sqrt{2}}( 0\rangle -  1\rangle) 1\rangle$ | $ \psi_{11}\rangle = \frac{1}{\sqrt{2}}( 01\rangle -  10\rangle)$ |

## Synthesis: The Dual Power of Quantum Architectures



### Classical Emulation

Reversible quantum gates (like Toffoli and CNOT) can perfectly simulate any classical boolean logic, proving quantum circuits can execute standard arithmetic.



### Quantum Supremacy

Introducing superposition (H-Gate) before conditional logic (CNOT) generates entanglement—a state with no classical equivalent, unlocking uniquely quantum computational power.

By mastering the discrete building blocks (Atoms & Molecules),  
we unlock the architecture of next-generation computation.